

Title: Literature review on the relationship between genetics and neural tube defect

This paper review papers which had been published on the relationship between gene and NTD using Pub med and Google scholars to search journals.

Despite decades of investigation, the etiology of NTDs remains to be clearly elucidated. One of the main reasons for this data gap is the lack of suitable models with which to study early developmental events directly in human embryos ([Wallingford, 2005](#); [Lei and Finnell, 2016](#)). Fortunately, recent technological breakthroughs in the use of neural tube organoids provide a novel three-dimensional model system that enables us to better understand the development of the human neural tube in an *in vitro* system. Ideally of course, it is best to test hypotheses in human cells, although this is not always possible. Animal models are very important for a better understanding of neural tube development as they can point to several candidate processes for study in the human NTDs etiology (Harris, M.J. 2010). By using animal models such as the mouse, chick, *Xenopus*, and zebrafish, investigators have successfully constructed multiple NTD models with which to probe at the cellular level the underlying mechanisms of failed neural tube closure ([Hildebrand and Soriano, 1999](#); [Gray et al., 2009](#)). While NTC in mouse embryos closely approximates human NTC, there are still some differences between mouse and human. Given these species-specific differences, the use of animal models has their limitations with respect to fully recapitulating the morphogenetic processes involved in human NTC. NTC in human embryos is initiated from only two closure sites, which is equivalent to the closure point 1 and 3 in mice of which there are four closure sites ([Golden and Chernoff, 1993](#)).

NTDs are believed to result from multiple factors with both genetic and environmental contributions ([Copp et al., 2013](#); [Wallingford et al., 2013](#); [Ross et al., 2017](#)). Human epidemiological evidence for a genetic component derives from the strong concordance of NTDs between mono-zygotic twins (7.7%) compared to like di-zygotic twins (4.0%; [Shaw et al., 1994](#); [Wallingford et al., 2013](#)). Furthermore, while neural tube closure defects can be familial after one NTD affected pregnancy, the recurrence risk is 1 in 20, and even after two affected pregnancies, the recurrence risk does not exceed 10%, strongly arguing against a monogenic causation. A number of case-control studies focusing on one, or at most, a few candidate genes have been used to identify alleles suggestive of an association with increased NTD risk. On the strength of folic acid NTD

prevention studies, the interrogation of a thermolabile variant (C677T) in the 5-MTHFR gene was proposed to increase the NTD risk ([Shields et al., 1999](#)) in some but not all, studied cohorts. Candidate genes selected either on the basis of mouse studies or trends in human genome sequences of relatively small case numbers have been replicated in human targeted re-sequencing studies for a number of genes, including many in the planar cell polarity ([Kibar et al., 2007](#); [Robinson et al., 2012](#)).

The well documented over 240 genes, whose mutation cause NTDs in the mouse (Harris and Juriloff, 2010), support the likelihood that numerous gene defects contribute to NTDs. Mouse genetic studies have also provided the insight that genetic background significantly effects the penetrance of NTDs in individual mice harboring those previously identified mutations and modifier loci that have been mapped in several mutant lines ([Juriloff et al., 2001](#); [Korstanje et al., 2008](#)).

As the association had been studied in some other papers, most null Murine mutants (>90%) present as fetuses with many affected developing organs with high penetrance in homozygotes, while some mutations cause NTDs in digenic, trigenic, and oligo genic combinations, an etiology that is consistent with the genetic causation described in human NTD patients ([Chen et al., 2018](#)), as human NTDs most often arise through an monogenic interplay of deleterious genetic variants and environmental factors influencing the function of core pathways such as OCM ([Boyle et al., 2017](#); [Chen et al., 2018](#)). What is important to note is that, in spite of strong data documenting, the role of genetic factors in the etiology of NTDs, there are no clinically actionable NTD candidate genes known at this time that influence the management of high risk pregnancies. The advent of next generation sequencing (NGS) opens up greater possibilities of dissecting out the genomic architecture underlying NTDs in the coming years.

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